

11.1) Hrabce singularny rozklad matice A :
 $AA^T = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 2 & 2 \\ -2 & 2 & 3 \end{pmatrix}$ je symetricka a vlastní čísla jsou 9, 9, 3 a vlastní vektory $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.
 Pak $A = UDV^T$, kde $U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$ a $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Dopíšeme V pomocí $v_i = \frac{1}{\sigma_i} A^T u_i$ a vyjde:

$$V = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

Matice singularny rozklad matice A .

Matice $A = UDV^T$, kde $A^T = VD^T U^T$.

Potom $AA^T = UDV^T VD^T U^T = UDD^T U^T$ kde $DD^T = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$.

$A^T A = VD^T U^T UDV^T = VD^T DV^T$ kde $DD^T = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$.